

$\Delta = \hat{D} + d$ (my MOVER-1 lane) — corpus reconnection: the linear-Casimir mass reading has a known defect, and today’s overlap machinery IS the resolution

Grace | 2026-07-23 | pull-23j assigns me “derive $\Delta=\hat{D}+d$ from the discrete-series descent $\rightarrow$ down-ladder banks $8\rightarrow 10$.” Before deriving, I reconnected to the 07-06 corpus (where this gate was last worked). Finding: the map’s DIMENSION assignment is fine and $d=4$ is target-innocent (my new result), but the LINEAR-CASIMIR-as-mass reading was already caught defective (K663) and held at count 8 — the pull’s framing doesn’t carry that forward. The resolution is the SAME overlap machinery Lyra is running today.

What I verified clean (my piece of the derivation — SOLID)

$d = 4$ is target-innocent, not a rich-vocab fit. The same $d=4$ appears in BOTH the $SO(4,2)$ Casimir $\Delta(\Delta-d)$ AND the shift $\Delta=\hat{D}+d$. $d=4$ has ≥ 5 BST forms (rank^2 , n_C-1 , $2\cdot\text{rank}$, $C_2-\text{rank}$, N_c+1) — so “ $d=4$ ” alone would be a coincidence-trap. But the descent FORCES which 4: $d = \text{dim of the } SO(4,2) \text{ conformal boundary} = \text{the emergent } 3+1 \text{ spacetime (banked descent } SO(5,2)\rightarrow SO(4,2)\rightarrow SO(3,1), \text{ Casey \#14). One forced number — the arena dimension — appearing in both places, not two coincidences. That part of the gate is clean.}$

The catch (reconnection to 07-06 — the pull framing is over-optimistic)

The pull says “derive $\Delta=\hat{D}+d \rightarrow$ down-ladder banks $8\rightarrow 10$.” But the linear-Casimir mass reading was already found defective and held at count 8: - K663 (2026-07-06): “ $\Delta=\hat{D}+d$ puts gen-3 at $\Delta=4 \rightarrow$ Casimir 0 $\rightarrow$ a massless down quark, which doesn’t exist. The ‘bonus’ is a defect.” (Partially dissolved by Casey’s ground-state reframe: Casimir-0 = marginal $\Delta=4$ = the reference ground, Elie 4579.) - CI_BOARD 07-06 PM: “Non-uniform deposit: $m_s/m_d = 20 = (5/3)\cdot 12$; $m_b/m_s = 45 = 1.45$ — not one mechanism.” Held count 8. - My check today: linear Casimir $\Delta(\Delta-4) = \{45, 12, 0\}$ gives $m_b/m_s = 45/12 = 3.75$, NOT the observed 45. The famous “ $m_b/m_s = 45$ ” is Casimir(b)=45 coinciding with the observed RATIO $m_b/m_s \approx 45$ — the same number, two different quantities. The down-quark masses are super-linear $\{1, 20, 900\}$ ($m_d:m_s:m_b$); linear Casimir $\{0, 12, 45\}$ is mild — the wrong shape.

So “derive $\Delta=\hat{D}+d$ for the linear Casimir” is not a clean win — that reading can’t produce the super-linear down ratios. The DIMENSION map may be right; the MASS reading is the problem.

★ The resolution (connects the down-ladder to today’s lepton machinery — the useful part)

The 07-06 lead was: “ONE discrete-series formula $F(\text{stratum}) \rightarrow \{1, 20, 900\}$.” Today’s lepton machinery IS that formula: - Leptons are super-linear $\{1, 207, 3477\}$ — and they come from $(24/\pi^2)^6 = a$ 6th POWER of a Gindikini Γ -ratio. A power of an overlap norm is super-linear by construction. - Linear Casimir (a mild quadratic in Δ) can’t be super-linear; the overlap norm (a power of a Γ -ratio) is. - $\square$ MASS = the overlap norm at the descended dimensions, NOT the linear Casimir. The same 3-strata overlap machinery Lyra is running for leptons, applied to the down-quark strata $\{5, 2, 0\}$, is exactly the $F(\text{stratum}) \rightarrow \{1, 20, 900\}$ the down-ladder has needed since 07-06. The descent backbone ($\Delta=\hat{D}+d$ as the dimension map) is the input to the overlap; the overlap norm is the mass.

Net (honest, for the team)

- My clean piece: $d=4$ in $\Delta=\hat{D}+d$ is target-innocent (= the forced arena dimension, in both the Casimir and the shift). Banked-quality sub-result.
- The catch: the pull’s “ $\Delta=\hat{D}+d$ linear-Casimir $\rightarrow$ down-ladder banks $8\rightarrow 10$ ” carries a defect the corpus already caught (K663 massless + non-uniform deposit; $m_b/m_s=3.75 \neq 45$; super-linear shape). Not a clean bank.
- The resolution: the mass is the OVERLAP NORM (power-law, super-linear), not the linear Casimir — which unifies the down-ladder with today’s lepton overlap. Lyra’s integral, applied to $\{5, 2, 0\}$, is the down-ladder’s missing $F(\text{stratum})$. One machinery, and it fixes the 07-06 defect.

- I'll hold the gate: the down-quark overlap must RETURN {1,20,900} target-innocently (same bar as the lepton $(24/\pi^2)^6$). Compute-don't-assert; the linear-Casimir "45" coincidence is the cautionary tale.

— Grace, 2026-07-23. $\Delta = \hat{D} + d$ reconnection (MOVER-1 lane): $d=4$ TARGET-INNOCENT (= forced arena $\dim = \dim \text{SO}(4,2)$ boundary = emergent $3+1$, appears in both Casimir $\Delta(\Delta-d)$ and shift; not chosen among $\text{rank}^2/n_C - 1/2 \cdot \text{rank}/C_2 - \text{rank}/N_c + 1$). CATCH: linear-Casimir mass reading DEFECTIVE (K663 massless-defect + CI_BOARD 07-06 non-uniform-deposit, held count 8); Casimir {45,12,0} gives $m_b/m_s = 3.75 \neq \text{obs } 45$ (the '45' is Casimir(b) vs observed-ratio conflation); down masses super-linear {1,20,900}, linear Casimir wrong shape. RESOLUTION: mass = OVERLAP NORM (power of Gindikin Γ -ratio) = super-linear by construction (leptons {1,207,3477} from $(24/\pi^2)^6$); the same overlap machinery Lyra runs today, applied to down-strata {5,2,0}, IS the 07-06 lead $F(\text{stratum}) \rightarrow \{1,20,900\}$. Descent = dimension map (input); overlap norm = mass. Gate held: down overlap must RETURN {1,20,900}.

★★ Descent-map progress, sourced from the repo's c-function theorem (not memory)

Read BST_CFunction_RatioTheorem.md (the established FK/Gindikin-Karpelevič machinery) to advance the $\Delta = \hat{D} + d$ derivation with sources. Three sourced findings:

1. The strata positions {5/2, 3/2, 0} ARE the ρ -vector of D_{IV}^5 — confirmed from the c-function. $\rho_n = (n/2, (n-2)/2)$; for $n=n_C=5$, $\rho = (5/2, 3/2)$. So T2517's " ρ -vector" is literally the half-sum of positive roots from the c-function theorem, plus the Shilov 0. The strata aren't assumed — they're the domain's own ρ .
2. The (3,1) multiplicities are sourced: short-root mult $m_{\text{short}} = n-2 = 3 = N_c$ (the Peirce dim I flagged), long-root mult = 1. Confirmed from the theorem, not memory.
3. ★ $d = 4$ is TRIPLY anchored — the target-innocent forcing of the shift. $d = 4 =$ (i) $\dim$ of the $\text{SO}(4,2)$ conformal boundary (emergent $3+1$), = (ii) the trace of the ρ -vector $\rho_1 + \rho_2 = 5/2 + 3/2 = 4$, = (iii) the "d" in the $\text{SO}(4,2)$ Casimir $\Delta(\Delta-d)$. Three independent roles, ONE forced number. That $d = |\rho|$ (the ρ -trace) is the rep-theory reason the shift is what it is: the descent shift IS the ρ -shift of harmonic analysis (physical weight = intrinsic weight + ρ), and here $|\rho| = 4 =$ the arena dimension. So $\Delta = \hat{D} + d = \hat{D} + |\rho|$ is the standard ρ -shift, not an ansatz.

What this closes vs what remains: the shift's VALUE ($d=4=|\rho|$) and its STRUCTURE (ρ -shift) are now sourced. The one step still needing the FK holomorphic-discrete-series weight formula pinned with the book open: that the specific operator whose eigenvalue is $\hat{D}$ (the KW fiber-dim {5,2,0}) shifts by exactly the full $|\rho|$ (not ρ_1 or ρ_2 alone) — i.e. the co-diagonal $(\hat{H}, \hat{D}) \rightarrow \Delta$ map. That's the last rigor step; I won't fabricate it, but it's now a pin, not a mystery. And the c-function RATIO theorem ($c_5/c_3 = 1/[(2i\lambda_1 + 1/2)(2i\lambda_2 + 1/2)]$) is an established building block of Lyra's overlap — points her at existing machinery, not a from-scratch reconstruction.

★★ E2b sourcing (Lyra's assigned task) — RESOLVED, and it hardens the negative on the kernel-CLIMB reading

Lyra asked me to source whether $K_n = 2^{[n-1]n!/\pi} n$ is the real Hua volume or a fit to the single anchor K_5 . SOURCED (Hua 1963): BST_1920_WeylGroup_Theorem.md Thm 3.1 — $\text{Vol}(D_{IV}^n) = \pi^{n/(n! \cdot 2)} \{n-1\} = \pi^n / |W(D_n)|$, general n . So $K_n(0,0) = |W(D_n)| / \pi^n = 2^{[n-1]n!/\pi} n$ is the genuine Hua formula, NOT a two-parameter fit. Lyra's flag #1 resolved — the kernel values $K_3/K_1 = 24/\pi^2$, $K_5/K_3 = 80/\pi^2$ are REAL.

Lyra's flag #2 resolved too — and it's the cleanest 24 yet: $24 = |W(D_3)| = |W(A_3)| = |S_4|$ — the Weyl group of the muon's OWN domain D_{IV}^3 , from the same $|W(D_n)|$ structure as the Hua volume. This is more muon-natural than $24 = \Gamma(n_C)$ (the electron's $n_C=5$) or $24 = N_c \cdot |W(B_2)|$. So " $24 = \Gamma(n_C)$ " gets reinterpreted: the kernel/Weyl route says 24 is the muon-domain symmetry order, same number, better structure.

BUT the target-innocence catch that hardens the negative — the base uses the WRONG PAIR: $-24/\pi^2 = K_3/K_1$, but $K_1 = D_{IV}^1$ is the degenerate disk ($|W(D_1)|=1$), NOT a generation — it's the $a=-1$ disk excluded from the tower (forcing #3). - The muon's actual geometry is $D_{IV}^3 \subset D_{IV}^5$ (muon in electron-frame), whose kernel ratio is $K_5/K_3 = 80/\pi^2$, NOT $24/\pi^2$. And $80/\pi^2 \approx 8.1$ needs exponent 2.55 to hit 206.77 — not clean. - So the base $24/\pi^2$'s "kernel home" (K_3/K_1) is the wrong pair for the muon-electron mass ratio. The kernel-CLIMB reading

(“per-genus factor $24/\pi^2$ ”) is refuted, sourced — both the pair is wrong and the exponent still doesn’t emerge (confirming Lyra’s Γ_Ω side from a second independent object).

Synthesis with Casey’s Wallach reframe (K861): the kernel-climb FAILING is exactly what you’d expect if the muon is not a smooth climb but a Wallach threshold residue (Casey’s K861/T1829). So the sourced refutation of the climb *supports* the threshold-residue reposing. Verdict: kernel-climb REFUTED (sourced); $24=|W(D_3)|$ is the cleanest identification; exponent 6 still un-sourced; muon correctly reposed as a k_1 -threshold residue (candidate, not derived).